



# Camber

## Structural Health Monitoring — Tester's Guide

Version 0.1.0 • Windows desktop • Everything runs on your PC — your data never leaves your computer.

Thank you for testing Camber. This guide walks you through installing the app, loading data, viewing and analysing sensor readings, and sending feedback. No prior setup or internet connection is required.

### 1. What Camber is

Camber is an open platform for **structural health monitoring (SHM)** — watching how a bridge (or other structure) behaves through data from its sensors: strain gauges, accelerometers, inclinometers, temperature and wind sensors, and more. With Camber you can **import a raw sensor recording** (a CSV export from a data logger), have the app **detect the layout automatically** and create the asset and its sensors, then **chart, monitor and analyse** the readings — including a **Live** view and an **FFT / spectrogram** for vibration analysis.

### 2. Installing & running

Camber is delivered as a standard **Windows installer**, `Camber_Setup.exe`.

**1.** Double-click `Camber_Setup.exe`. If Windows shows a "Windows protected your PC" (SmartScreen) notice on an unsigned build, click **More info** → **Run anyway**. It installs for your user only and does not need administrator rights.

**2.** Finish the wizard. You get a **Start Menu** entry and, if you ticked it, a **Desktop** shortcut.

**3.** Launch **Camber**. The main window opens; a small local service also starts in the background at `http://127.0.0.1:8765` (used only on this machine — see §11 Privacy).

**First run.** The app starts with an empty inventory. The fastest way to see it working is **Data** → **Load sample data** (§4).

### 3. The window at a glance

Camber is organised into tabs, plus a small menu bar (**Data** and **Help**) at the top.

| Tab              | What it shows                                                                              |
|------------------|--------------------------------------------------------------------------------------------|
| <b>Dashboard</b> | A summary of your assets and their overall status.                                         |
| <b>Assets</b>    | Your inventory of structures. Import files here, and <b>Rename</b> / <b>Delete</b> assets. |
| <b>Status</b>    | Per-sensor OK / Warning / Critical against threshold rules.                                |
| <b>Charts</b>    | Time-series plot of a chosen sensor, with a <b>Live</b> mode.                              |
| <b>Analysis</b>  | Frequency analysis — FFT spectrum and spectrogram (best on accelerometers).                |
| <b>Map</b>       | Asset locations on a map (when coordinates are known).                                     |

**Menu bar.** **Data** → **Load sample data** loads a bundled example recording. **Help** → **View logs folder** opens the folder with the app's log files, **Help** → **Report an issue** opens the feedback page, and **Help** → **About** shows the version and log location.

## 4. Quick start in 60 seconds

1. Open Camber. 2. Click the menu **Data** → **Load sample data**. A confirmation reports how many readings and sensors were loaded into "Vänernsberg Bridge (sample)".

3. Go to the **Charts** tab, pick a sensor (e.g. an accelerometer A1–A5), and click **Refresh**.

4. Tick **Live**, then click **Demo feed** to watch new readings stream in. 5. Open the **Analysis** tab and pick an accelerometer to see its frequency spectrum. That's the whole loop.

## 5. Importing your own sensor files

Camber reads **CSV** files. There are two buttons on the **Assets** tab.

### Import sensor file... (recommended)

Use this for a **raw recording from a data logger** — including a **wide**, multi-channel file where each column is a channel (ch\_1, ch\_2, ...). Camber:

- auto-detects the **delimiter** (comma, semicolon or tab), the **decimal mark** (., or ,), the text encoding and the timestamp format — so files from different countries and vendors import correctly;

- shows a **preview** of the detected layout before importing; and
- **creates the asset and its sensors for you**, so you don't set anything up by hand.

In the preview dialog you can review the detected format and warnings, then choose to import into a **new asset** (give it a name) or an **existing** one. Click **Import** and a progress dialog tracks the readings as they load.

### The import profile (for known loggers)

A **profile** is a small mapping that tells Camber what each column means (e.g. "`ch_1` = strain gauge SG1 in  $\mu\text{m/m}$ "). Camber ships with a profile for the Vänersborg-style 30-channel logger and matches it automatically. You can add your own profiles for a recurring file format — ask us and we'll help you write one.

### Import CSV (tidy files)

Use the plain **Import CSV** button when you already have a **tidy** measurements file with the columns `sensor_id`, `timestamp`, `metric_type`, `value`, `unit`, aimed at sensors that already exist. If you pick a raw multi-channel file here by mistake, Camber tells you to use **Import sensor file...** instead.

**Tip.** If an import ever fails, note the message and check **Help** → **View logs folder** — the full detail is written there, which helps us fix the file quickly.

## 6. Charts — viewing time series

On the **Charts** tab, choose a **Sensor** from the drop-down and click **Refresh**. The plot shows that sensor's readings over time. If a threshold rule exists for the metric, you'll see shaded **warning** and **critical** bands and the points coloured by status.

You can zoom (mouse wheel), pan (drag), and right-click the plot for view options.

Large recordings are drawn as a fast down-sampled line automatically.

## 7. Live mode

Tick the **Live** checkbox to make the chart **auto-refresh and follow the latest readings** as they arrive. This is how a real, always-on bridge sensor would appear.

To see Live mode without hardware, click **Demo feed**: Camber generates realistic synthetic readings for the selected sensor so you can watch the live chart move.

Turn it off to stop. (Connecting real sensors is covered in §10.)

## 8. Analysis — FFT & spectrogram

The **Analysis** tab turns a vibration signal into its **frequencies** — useful for finding a structure’s natural (modal) frequencies. Pick an **accelerometer**, and Camber computes:

- an **FFT amplitude spectrum** (how much energy sits at each frequency), and
- a **spectrogram** (how that spectrum changes over time).

The sampling rate is estimated automatically from the timestamps, so you normally don’t enter anything. Peaks in the spectrum correspond to the structure’s vibration modes.

## 9. Managing assets

On the **Assets** tab, select a row and use:

- **Rename** — give the asset a clearer name.
- **Delete** — remove the asset **and all its sensors and measurements**. You are asked to confirm; this cannot be undone.
- **Refresh** — reload the list.

## 10. Connecting real live sensors (advanced)

Camber has a small **local ingest API** that live readings are pushed to. It listens only on this PC at `http://127.0.0.1:8765`. The Demo feed and the replay tool both use it, and a real **sensor gateway** would use the same path.

**Push one reading** — send a JSON `POST` to `/measurements`:

```
{ "sensor_id": 12, "value": 0.83, "metric_type": "acceleration", "unit": "m/s2" }
```

**Push many at once** — `POST` a JSON list to `/measurements/batch`. Ticking **Live** on the Charts tab then shows them arriving.

**Replay a recording as if it were live** — a helper script plays an existing CSV back through the API at a chosen rate:

```
python scripts\replay_csv.py --list # list assets
python scripts\replay_csv.py RECORDING.csv --asset-id 3 # replay into asset 3
```

**Connecting actual hardware.** A “gateway” is a small program that reads from your data logger (over a folder it writes CSVs to, or a protocol such as MQTT / Modbus / a vendor SDK) and forwards each reading to the endpoint above. Camber does not

talk to a specific logger yet — tell us your hardware and we'll build the gateway for it.

## 11. Privacy — where your data lives

Camber is a **desktop application**. Your sensor data is stored in a local database file on your own computer and is **never uploaded anywhere**. There is no account, no cloud, and no website. The background service is bound to `127.0.0.1` (loopback), meaning it is reachable only from your own machine — not from the network or the internet.

## 12. Troubleshooting & feedback

- **Something looks wrong or the app shows an error dialog.** The app keeps running and writes details to a log file. Open **Help** → **View logs folder** and send us the newest `.log` file.
- **Report an issue.** Use **Help** → **Report an issue**, or email us, with: what you did, what you expected, what happened, and (if possible) the file you imported and the log.
- **An import failed.** Almost always a file-format detail — send the first few rows of the CSV and the log, and we'll add support for it.

## 13. Known limits

This is an early (v0.1) build for evaluation. It is well suited to **campaign and event data** — recordings and monitoring of bridges over hours, days or specific events. It is **not yet** built for permanently streaming very high-rate data (e.g. continuous 200 Hz from many channels forever); that path (columnar storage, streaming, automated modal analysis) is on the roadmap. If you hit a limit, that is useful feedback — please tell us.

Camber — open structural health monitoring. Local API: `http://127.0.0.1:8765` • This guide covers version 0.1.0.